schematically represent on the printed page by a two-dimensional grid). However, when we get down to the Planck scale, the highest level of magnification in the figure, Klein suggested that a new, curled-up dimension becomes visible.

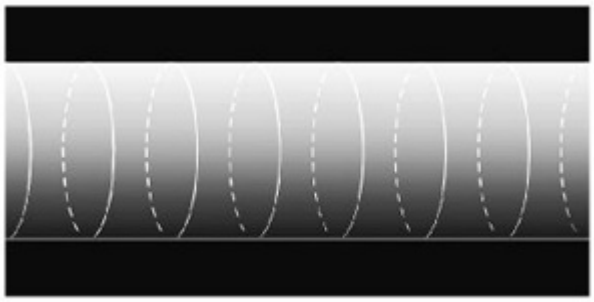


Figure 12.6 The surface of a tightrope has one long dimension with a circular dimension tacked on at each point.

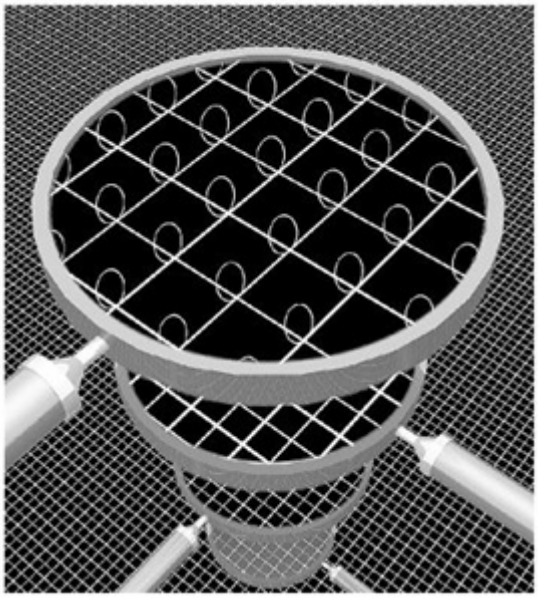


Figure 12.7 The Kaluza-Klein proposal is that on very small scales, space has an extra circular dimension tacked on to each familiar point.

Just as the circular dimension of the tightrope exists at each point along its big, extended dimension, the circular dimension in this proposal exists at each point in the familiar three extended dimensions of daily life. In Figure 12.7, we illustrate this by drawing the additional circular dimension at various points along the extended dimensions (since drawing the circle at every point would obscure the image) and you can immediately see the similarity with the tightrope in Figure 12.6. In